Nishal Stanislaus Silva, *PhD*

Research Scientist | Real-Time Multimodal ML, Benchmarks & Open Datasets

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Re: Research Scientist – Waabi

Dear Hiring Manager,

I am applying for the Research Scientist role on the Autonomy and Algorithms team at Waabi. I am a research scientist in real-time multimodal machine learning, with a PhD from the University of Trento and ten years across industry and academia. My subfield is not self-driving, and I want to be direct about that. What I bring is a decade of work on the problems underneath it: fusing heterogeneous sensor streams under hard timing constraints, modeling behavior with hierarchical state machines, generating synthetic data where labels are scarce, and proving that a model holds up with benchmarks rather than hope.

Concretely: during my postdoc I built a real-time pipeline that synchronized audio, inertial, neural and mixed-reality tracking data across four participants at once. At McGill I fused inertial and audio streams through a hierarchical state machine that moved between idle, pattern and gesture modes, reaching F1 0.78 on a 500-event corpus. My published detector runs inference in 14 ms on embedded hardware, 74 times faster than the baseline it replaced. I have designed four open datasets used as detection benchmarks and I review for IEEE venues. The transfer to perception, motion forecasting, behavior modeling and sensor simulation is real, and I would expect to spend my first months closing the gap on the autonomy literature while contributing on the sensor-fusion and evaluation side from day one.

Waabi is the reason I would make that move. The June result, transferring the Waabi Driver to a new vehicle platform with no retraining and no training examples, is exactly the kind of generalization I find most interesting, and it rhymes with the training-free and zero-shot methods in my own work. A single end-to-end system that learns and is provably safe, backed by Waabi World, is a research bet worth being part of, and it is based in Toronto.

I am a Canadian permanent resident based in Toronto, eligible to work without sponsorship and available immediately. I would welcome the opportunity to discuss how my background in real-time multimodal ML can support the Autonomy & Algorithms team at Waabi. My publications, datasets and portfolio are at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD